CodeSignal Test 5

Practice Questions

Total: 62 Questions | Correct answers marked in green

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1. A developer plans to launch an EC2 instance, with Amazon Linux 2 as its AMI, using the AWS Console. A security group with port 80 that is open to public access will be associated with the instance. He wants to quickly build and test an Apache webserver with an index.html displaying a hello world message. Which of the following should the developer do?

A. Configure the metadata at the creation of the EC2 Instance to run a script that will install the Apache webserver after the instance starts.

B. Connect to the instance via port 80. Run the commands that will install and create the Apache webserver.

C. Connect to the instance via port 22. Run the commands that will install and create the Apache webserver.

D. Configure the user data at the creation of the EC2 Instance to run a script that will install and create the Apache webserver after the instance starts.

2. A developer is hosting a static website from an S3 bucket. The website makes requests to an API Gateway endpoint integrated with a Lambda function (non-proxy). The developer noticed that the requests were failing. Upon debugging, he found a "No 'Access-Control-Allow-Origin' header is present on the requested resource" error message. What should the developer do to resolve this issue?

A. Set the value of the Access-Control-Man-Age header to 0.

B. In the Amazon S3 Console, enable cross-origin resource sharing (CORS) on the S3 bucket where the website is hosted.

C. Set the value of the Access-Control-Allow-Credentials header to true.

D. In the API Gateway Console, enable cross-origin resource sharing (CORS) for the method in the specified resource.

3. A San Francisco-based tech startup is building a cross-platform mobile app that can notify the user of upcoming astronomical events. Your mobile app authenticates with the Identity Provider (IdP) using the provider's SDK and Amazon Cognito. Once the end-user is authenticated with the IdP, the (Auth or OpenId) Connect token returned from the IdP is passed by your app to Amazon Cognito. Which of the following is returned for the user to provide a set of temporary, limited-privilege AWS credentials?

A. Cognito SDK

B. Cognito API

C. Cognito Key Pair

D. Cognito ID

4. A company is planning to launch an online cross-platform game that expects millions of users. The developer wants to use an in-house authentication system for user identification. Each user identifier must be kept consistent across devices and platforms. How can the developer achieve this?

A. Use developer-authenticated identities in Amazon Cognito to generate unique identifiers for the users.

B. Create an IAM Role for each user and use its Amazon Resource Name (ARN) as unique identifiers.

C. Generate a universally unique identifier (UUID) for each device. Store the UUID with the user in a DynamoDB table.

D. Generate a unique IAM access key for each user and use the access key ID as the unique identifier.

5. A developer wants to cut down the execution time of the scan operation on a DynamoDB table during periods of low demand without interfering with typical workloads. The operation consumes half of the strongly consistent read capacity units within regular operating hours. How can the developer improve this scan operation?

A. Perform a rate-limited sequential scan operation.

B. Perform a rate-limited parallel scan operation.

C. Use a parallel scan operation.

D. Use eventually consistent reads for the scan operation instead of strongly consistent reads.

6. A development team uses AWS Step Functions to orchestrate a serverless workflow that processes incoming data streams with AWS Lambda. The team requires a solution for capturing custom metrics, such as processing times, directly from the logs generated by an AWS Lambda function. These metrics must be analyzed for operational insights, with alarms set up to detect anomalies in real-time. Which approach should be used to meet this requirement?

A. Use Amazon's open-source libraries to format logs in the Amazon CloudWatch embedded-metric format. Then, use CloudWatch to monitor, view, and set alarms for the custom metrics.

B. Send custom metric data directly to Amazon EventBridge using the PutMetricData API operation. Create EventBridge rules to trigger actions based on the metrics.

C. Use CloudWatch Lambda Insights to extract and monitor custom metrics for processing times. Set up alarms and dashboards in Lambda Insights for real-time analysis.

D. Configure Amazon Data Firehose to stream logs from the Lambda function to an Amazon Redshift cluster. Use SQL queries to analyze metrics and set alerts for anomalies.

7. A developer is building an application with Amazon DynamoDB as its database. The application needs to group the PUT, UPDATE, and DELETE actions into a single all-or-nothing operation to make changes against multiple items in the DynamoDB table. Which DynamoDB operation should the developer use?

A. Scan

B. BatchWriteItem

C. TransactWriteItems

D. Query

8. A developer has enabled the lifecycle policy of an application deployed in Elastic Beanstalk. The lifecycle is set to limit the application version to 10 versions. The developer wants to keep the source code in an S3 bucket, yet, it gets deleted. What change should the developer do?

A. Modify the value of the Set application versions limit by the total count option to zero.

B. Modify the value of the Set the application versions limit by age option to zero.

C. Trigger a Lambda function to copy the source code to another S3 bucket.

D. Configure the Retention setting to retain the source bundle in the S3 bucket.

9. A developer wants to expose a legacy web service that uses an XML-based Simple Object Access Protocol (SOAP) interface through API Gateway. However, there is a compatibility issue since most modern applications communicate data in JSON format. Which is the most cost-effective method that will overcome this issue?

A. Use API Gateway to create a RESTful API. Send the incoming JSON to an HTTP server hosted on an EC2 instance and have it transform the data into XML and vice versa before sending it to the legacy application.

B. Use API Gateway to create a RESTful API. Transform the incoming JSON into XML using a Lambda function and parse the response (XML) into JSON before sending back to API Gateway.

C. Use API Gateway to create a WebSocket API. Transform the incoming JSON into XML using mapping templates. Forward the request into the SOAP interface by using a Lambda function and parse the response (XML) into JSON before sending back to API Gateway.

D. Use API Gateway to create a RESTful API. Transform the incoming JSON into XML for the SOAP interface through an Application Load Balancer and vice versa. Put the legacy web service behind the ALB.

10. An application has a feature that displays GIFs based on keyword inputs. The code streams random GIF links from an external API to your local machine. When run, the application's process takes longer than expected. You are suspecting that the new function sendRequest() you added is the culprit. Which of the following actions should you do to determine the latency of the function?

A. Using CloudWatch, troubleshoot the issue by checking the logs.

B. Use CloudTrail to record and store event logs for actions made by your function.

C. Using AWS X-Ray, disable sampling to efficiently trace all requests for calls.

D. Using AWS X-Ray, define an arbitrary subsegment inside the code to instrument the function.

11. A company has a serverless application on AWS. They are using AWS Lambda for business logic and Amazon API Gateway for handling client requests. The internal team has published a version of the AccountService:Prod function with the alias AccountService:Beta. The internal team wants to test these updates before promoting them to production without impacting live users. Which configuration should the company take?

A. Create a new stage named 'Beta' in Amazon API Gateway and use stage variables to reference the Lambda functions in Prod and Beta.

B. Create a 'Beta' stage in Amazon API Gateway in the Lambda function configuration, add a condition to the AccountService:Beta alias that checks for an environment variable and, if set to 'Beta', invokes the AccountService:Beta alias instead.

C. Update the integration request settings in the 'Prod' stage of the Amazon API Gateway to select the Lambda function alias based on a query parameter.

D. Modify the existing 'Prod' stage in Amazon API Gateway to create a new stage variable that references the Lambda functions in Prod and Beta.

12. A development team needs to deploy an application revision into three environments: Test, Staging, and Production. The application should be deployed into the Test environment first, then Staging, and then Production. Which approach will conveniently allow the team to deploy the application into different environments?

A. Create multiple deployment groups for each environment using AWS CodeDeploy.

B. Create, configure, and deploy multiple application projects for each environment using CodeBuild.

C. Create separate CloudFormation templates for each environment to deploy the application.

D. Create separate S3 buckets for each environment to deploy the application.

13. Some static assets stored in an S3 bucket need to be accessed by a user on the development account. The S3 bucket is in the production account. According to the company, the sharing of full credentials between accounts is prohibited. What steps should be done to delegate access across the two accounts? (Select THREE.)

A. Set the policy that will grant access to S3 for the IAM role created in the production account.

B. Log in to the production account and create a policy that will use STS to assume the IAM role in the development account. Attach the policy to the IAM users.

C. Log in to the development account and create a policy that will use STS to assume the IAM role in the production account. Attach the policy to the IAM users.

D. Set the policy that will grant access to S3 for the IAM role created in the development account.

E. On the development account, create an IAM role and specify the production account as a trusted entity.

F. On the production account, create an IAM role and specify the development account as a trusted entity.

14. A developer is managing several microservices built using API Gateway and AWS Lambda. The Developer wants to deploy new updates to one of the APIs. He wants to ensure a smooth transition between the versions by giving users enough time to migrate to the new version before retiring the previous one. Which solution should the developer implement?

A. Implement the updates and publish a new version of the Lambda function. Specify the new version in the API Gateway target resource then redeploy it to a new stage.

B. Implement the updates and publish a new version of the Lambda function. Then, issue the new Lambda invocation URL.

C. Implement the updates on the Lambda function. Create a CloudFront distribution and use the function as the origin.

D. Implement the updates and publish a new version of the Lambda function. Specify the new version in the API Gateway target resource then redeploy it to the same stage.

15. A developer is building a serverless URL shortener using Amazon API Gateway, Amazon DynamoDB, and AWS Lambda. The application code as well as the stack that defines the cloud resources should be written in Python. The code should also be reusable in case an update must be done to the stack. Which of the following actions must be done by the developer to meet the requirements above?

A. Use AWS CDK to build the stack. Then, use Python as the runtime environment in writing the application logic on Lambda.

B. Use AWS CloudFormation to build the stack. Then, use Python as the runtime environment in writing the application logic on Lambda.

C. Use AWS CloudShell to build the stack. Then, use Python as the runtime environment in writing the application logic on Lambda.

D. Use AWS SDK for Python (boto3) to build the stack. Then, use Python as the runtime environment in writing the application logic on Lambda.

16. A developer needs to build a queuing mechanism for an application that will run on AWS. The application is expected to consume SQS messages that are larger than 256 KB and up to 1 GB in size. How should the developer manage the SQS messages?

A. Use Amazon S3 and the Amazon SQS Console

B. Use Amazon S3 and the Amazon SQS CLI

C. Use Amazon S3 and the Amazon SQS Extended Client Library for Java

D. Use Amazon S3 and the Amazon SQS HTTP API

17. A developer is testing a Lambda function that was created from a CloudFormation template. While the function executes without errors, it isn't generating logs in Amazon CloudWatch Logs, and the developer cannot find associated log streams or log groups. Upon inspection, the following observations were made: - The Lambda function is associated with an execution role that establishes a trusted relationship with the Lambda service; however, this role has no permissions. - The Lambda function does not have any resource-based policies. Which configuration must be done to resolve the issue?

A. Update the execution role by adding the CloudWatchLambdaInsightsExecutionRolePolicy managed policy.

B. Associate the function with a resource-based policy that contains the logs:PutLogEvents permissions.

C. Associate the function with a resource-based policy that contains the logs:CreateLogGroup, logs:CreateLogStream, and logs:PutLogEvents permissions.

D. Update the execution role by adding the AWSLambdaBasicExecutionRole managed policy.

18. A team is deploying a serverless app with AWS Cloud Development Kit (CDK), using the Lambda construct library to define Lambda functions. They've had successful deployments in a test environment using the cdk deploy command. However, when the team attempts their first deployment to a new AWS account, it fails, returning a NoSuchBucket error. What AWS CDK CLI command should be run first to fix the problem?

A. cdk synth

B. cdk context

C. cdk bootstrap

D. cdk import

19. A developer has an application that stores sensitive data to an Amazon DynamoDB table. AWS KMS must be used to encrypt the data before sending it to the table and to manage the encryption keys. Which of the following features are supported when using KMS? (Select TWO.)

A. Automatic key rotation for KMS keys in custom key stores

B. Re-enabling disabled keys

C. Creation of symmetric encryption and asymmetric KMS keys

D. Importing a custom key material to an asymmetric KMS key

E. Using AWS Certificate Manager as a custom key store

20. A developer is managing an Application Load Balancer that targets a Lambda function. The developer needs to obtain all values of identical query parameters key that is supplied in a request. How can the developer implement this?

A. Set a custom HTTP response header in the Lambda function.

B. Decode the URL-encoded query string values in the Lambda function.

C. Replace the Application Load Balancer with a Network Load Balancer and enable multi-value headers.

D. Enable the multi-value headers on the Application Load Balancer.

21. A company is running an e-commerce application on an Amazon EC2 instance. A newly hired developer has been tasked to monitor and handle necessary updates on the EC2 instance every Saturday. The developer is working from home and needs remote access to the webserver. As the system administrator, you're looking to use the AWS STS API to protect specific programmatic calls against the MFA device that could adversely affect the server. Which of the following STS API should you use?

A. GetFederationToken

B. AssumeRoleWithSAML

C. GetSessionToken

D. AssumeRoleWithWebIdentity

22. An application uses the PutObject operation in parallel to upload hundreds of thousands of objects per second to an S3 bucket. To meet security compliance, the developer uses the server-side encryption in AWS KMS (SSE-KMS) to encrypt objects as they get stored in the S3 bucket. There is a noticeable performance degradation after making the change. Which of the following is the most likely cause of the problem?

A. The AWS KMS key uses an AES 256 algorithm, which makes the encryption process slower. AES 128 should be used instead.

B. The Amazon S3 throttles the PutObject operation for objects encrypted with SSE-KMS.

C. The KMS key does not use an alias required for the server-side encryption with AWS KMS (SSE-KMS).

D. The API request rate has exceeded the quota for AWS KMS API operations.

23. A media analytics company provides APIs as a service that enables users to retrieve aggregated, daily-updated data on video performance metrics, such as views, likes, and watch times. These APIs are built using Amazon API Gateway and AWS Lambda, with data sourced from a pre-computed file in Amazon S3 that updates every 24 hours. Due to a surge in traffic, some users have reported latency. The company aims to enhance the API's responsiveness without changing the backend architecture. Which approach would improve the responsiveness of the APIs?

A. Enable cross-origin resource sharing (CORS) using the API Gateway

B. Use Amazon ElastiCache to store frequently-requested data in memory

C. Configure Amazon CloudFront as a caching layer in front of API Gateway

D. Enable Amazon API Gateway caching

24. A university is gradually migrating some of its physical documents to the AWS cloud. They will start by moving their alumnus' historical records to Amazon S3. The storage solution should provide a secure and durable object storage with the lowest cost. Which of the following types of S3 storage should you recommend?

A. Amazon S3 Glacier Deep Archive

B. Amazon S3 Infrequent Access

C. Amazon S3 Glacier

D. Amazon S3 One-Zone

25. An application executes GET operations to various AWS services. The development team is using AWS X-Ray to trace all the calls made to the application. To save time, you only want to record data associated with the code to group the traces in the AWS console. Which of the following X-Ray features should you use?

A. Subsegment

B. Metadata

C. Annotations

D. Sampling

26. An application is used to upload images to an Amazon S3 bucket. Once an event occurs, a Lambda function is triggered to compress the photos. However, it has been discovered that the processing time of the function is longer than expected. Which change will improve the processing time of the function most effectively?

A. Run the function with Lambda@Edge which will run the code closer to the users of your application, reducing your application's latency.

B. Configure the S3 bucket to send notifications to an SQS queue. Use the SQS queue with the Lambda function to process the image.

C. Increase the timeout value of the function.

D. Increase the memory allocation of the function.

27. A developer is writing a web application that will allow users to save and retrieve images in an Amazon S3 bucket. The users are required to register and log in to access the application. Which combination of AWS Services should the Developer utilize for implementing the user authentication module of the application?

A. Amazon Cognito User Pools and AWS Key Management Service (KMS)

B. Amazon User Pools and AWS Security Token Service (STS)

C. Amazon Cognito Identity Pools and IAM Role.

D. Amazon Cognito Identity Pools and User Pools.

28. A development team is building a website that displays an analytics dashboard. The team uses AWS CodeBuild to compile the website from a source code residing on Github. A member was instructed to configure CodeBuild to run with a proxy server for privacy and security reasons. A RequestError timeout error appears on CloudWatch whenever CodeBuild is accessed. Which is a possible solution to resolve the issue?

A. Modify the artifacts element of the buildspec.yml file on the source code root directory.

B. Modify the proxy element of the buildspec.yml file on the source code root directory.

C. Modify the proxy element of the AppSpec.yml file on the source code root directory.

D. Modify the phases element of the AppSpec.yml file on the source code root directory.

29. Several development teams worldwide will be collaboratively working on a project hosted on an AWS Elastic Beanstalk environment. The developers need to be able to deploy incremental code updates without re-uploading the entire project. Which of the following actions will reduce the upload and deployment time with the LEAST amount of effort?

A. Host the code repository on an EC2 instance and allow access to all the developers. Write a script that will automate the deployment process to Elastic Beanstalk.

B. Configure event notifications on a central S3 bucket and allow access to all developers. Invoke a Lambda Function that will deploy the code to Elastic Beanstalk when a PUT event occurs.

C. Upload the code to an Amazon EFS mounted on an EC2 instance. Write a script that will automate the deployment process to Elastic Beanstalk.

D. Create an AWS CodeCommit repository and allow access to all developers. Deploy the code to Elastic Beanstalk.

30. A developer is building an AWS Lambda-based Java application that optimizes pictures uploaded to an S3 bucket. Upon running several tests, the Lambda function shows a cold start of about 5 seconds. Which of the following could the developer do to reduce the cold start time? (Select TWO.)

A. Reduce the deployment package's size by including only the needed modules from the AWS SDK for Java.

B. Add the Spring Framework to the project and enable dependency injection.

C. Run the Lambda function in a VPC to gain access to Amazon's high-end infrastructure.

D. Increase the timeout setting for the Lambda function.

E. Increase the memory allocation setting for the Lambda function.

31. A developer has a Python script that relies on the low-level BatchGetItem API to fetch large amounts of data from a DynamoDB table. The script often encounters responses with partial results. A significant portion of the data appears under UnprocessedKeys. Which approaches can the developer implement to handle data retrieval MOST reliably? (Select TWO.)

A. Use the AWS software development kit (AWS SDK) to send batch requests.

B. Create a Global Secondary Index (GSI) with its own read capacity settings.

C. Implement a logic that immediately retries the batch request.

D. Increase the read capacity units (RCUs) for the DynamoDB table and enable Auto Scaling.

E. Implement an exponential backoff algorithm with a randomized delay between retries of the batch request.

32. A developer plans to use AWS Elastic Beanstalk to deploy a microservice application. The application will be implemented in a multi-container Docker environment. How should the developer configure the container definitions in the environment?

A. Use the eb config command to configure the container definitions.

B. Configure the container definitions in the Dockerrun.aws.json file.

C. Configure the container definitions in the Amazon ECS Console when building the Docker environment.

D. Configure the container definitions in the Dockerrun.aws.json.config and put it inside the .ebextensions folder.

33. An application is hosted in the us-east-1 region. The app needs to be recreated on the us-west-2, ap-northeast-1, and ap-southeast-1 regions using the same Amazon Machine Image (AMI). As the developer, you have to use AWS CloudFormation to rebuild the application using a template. Which of the following is the most suitable way to configure the CloudFormation template for this scenario?

A. Copy the AMI of the instance from the us-east-1 region to the us-west-2, ap-northeast-1, and ap-southeast-1 region. Then, add a Mappings section wherein you will define the different Image Id for the three regions. Use the region name as the key in mapping to its correct Image Id. Lastly, use the Fn::FindInMap function to retrieve the desired Image Id from the region key.

B. Copy the AMI of the instance from the us-east-1 region to the us-west-2, ap-northeast-1, and ap-southeast-1 region. Then, add a Mappings section wherein you will define the different Image Id for the three regions. Use the region name as the key in mapping to its correct Image Id. Lastly, use the Fn::GetAtt function to retrieve the desired Image Id from the region key.

C. Copy the AMI of the instance from the us-east-1 region to the us-west-2, ap-northeast-1, and ap-southeast-1 region. Then, add a Mappings section wherein you will define the different Image Id for the three regions. Use the region name as the key in mapping to its correct Image Id. Lastly, use the Fn::ImportValue function to retrieve the desired Image Id from the region key.

D. Copy the AMI of the instance from the us-east-1 region to the us-west-2, ap-northeast-1, and ap-southeast-1 region. Then, add a Parameters section wherein you will define the different Image Id for the three regions. Use the region name as the key in mapping to its correct Image Id. Lastly, use the Ref function to retrieve the desired Image Id from the region key.

34. A startup plans to use Amazon Cognito User Pools to easily manage their users' sign-up and sign-in workflows to an application. To save time from designing the User Interface (UI) for the Cognito's built-in UI, the development team has decided to include the product logo on the web page. However, the product manager finds out the UI and instructed the development team to include the product logo on the custom login page. How should the developer meet the above requirements?

A. Create a login page with the product logo and upload it to an S3 bucket. Point the S3 endpoint in the Cognito app settings.

B. Create a login page with the product logo and upload it to Amazon Cognito.

C. Upload the logo to an S3 bucket and point the S3 endpoint on the custom login page.

D. Upload the logo to the Amazon Cognito app settings and use that logo on the custom login page.

35. A serverless application is created to process numerous files. Each invocation takes 5 minutes to complete. The Lambda function's execution time is too slow for the application. Considering that the Lambda function does not return any important data, which method will accelerate data processing the most?

A. Compress the files to reduce their size, then process them with synchronous RequestResponse Lambda invocations.

B. Merge all of the files into a single file then process them all at once with an asynchronous Event Lambda invocation.

C. Use synchronous RequestResponse Lambda invocations. Process the files one by one.

D. Use asynchronous Event Lambda invocations. Configure the function to process the files in parallel.

36. A developer is building a serverless workflow using Step Functions. The developer has to implement a solution that will help the State Machine handle and recover from State exception errors. Which of the following actions should the developer do?

A. Use Catch and Retry fields inside the application code.

B. Use a try and catch logic inside the application code to capture the exception error. Then, use a Retry field in the state definition.

C. Use a Catch field inside the application code to capture the exception error. Then, use a Retry field in the state definition.

D. Use Catch and Retry fields in the state machine definition.

37. A developer is building a serverless API composed of an API Gateway and several Lambda functions. All resources are defined using AWS CDK L2 constructs. The developer wants to test some of the Lambda functions in their local environment. Given that AWS SAM and AWS CDK are already configured, what combination of actions must the developer do? (Select TWO.)

A. Run the cdk synth command and indicate the stack name of Lambda functions to be tested.

B. Execute the sam local invoke command and specify the location of the synthesized CloudFormation template and identifier of each function.

C. Execute the sam local start-lambda and specify the location of the synthesized CloudFormation template along with function identifiers.

D. Execute the sam package and specify the S3 bucket where the Lambda code will be uploaded for local testing.

E. Run the cdk bootstrap command to prepare the staging of CDK assets.

38. A company wants to know how its monolithic application will perform on a microservice architecture. The Lead Developer has deployed the application on Amazon ECS using the EC2 launch type. He terminated the container instance after testing; however, the container instance still appears as a resource in the ECS cluster. What is the possible cause of this?

A. After terminating the container instance in the stopped state, the container instance must be manually deregistered in the Amazon ECS Console since it was launched using the EC2 launch type.

B. After terminating the container instance in the running state, the container instance must be manually deregistered in the Amazon ECS Console.

C. When a container instance is terminated in the running state, the container instance is not automatically deregistered from the cluster.

D. When a container instance is terminated in the stopped state, the container instance is not automatically deregistered from the cluster.

39. A Microservices application's Customer and Payment service components have two separate DynamoDB tables. New items inserted into the Customer service table must be dynamically updated in the Payment service table. How can the Payment service get near real-time updates?

A. Create a Firehose stream to stream all the changes from the Customer service table and trigger a Lambda function to update the Payment service table.

B. Use a Kinesis data stream to stream all the changes from the Customer service database directly into the Payment service table.

C. Create a scheduled Amazon EventBridge (Amazon CloudWatch Events) rule that invokes a Lambda function every minute to change from the Customer service table into the Payment service table.

D. Enable DynamoDB Streams to stream all the changes from the Customer service table and trigger a Lambda function to update the Payment service table.

40. A Software Engineer is developing a Node.js application that will be deployed using Elastic Beanstalk. The application source code is currently inside a folder called MyApp. He wants to add a configuration file named tutorial:debug.config to the application. Where should the file be placed?

A. Inside the MyApp folder at the root level

B. Inside the .ebextensions folder

C. Inside the package.json

D. Inside the .elasticbeanstalk folder

41. A developer needs to use IAM roles to list all EC2 instances that belong to the development environment in an AWS account. Which methods could be done to verify IAM access to describe instances? (Select TWO.)

A. Run the get-role command.

B. Run the describe-instances command with the --dry-run parameter.

C. Validate the IAM role's permission by querying the in-line policies within the EC2 instance metadata.

D. Run the get-group-policy command.

E. Use the IAM Policy Simulator to validate the permission for the IAM role.

42. A mobile game developer is using DynamoDB as a data store and a Web Identity Federation for authorization and authentication. Each item in the DynamoDB table contains the attributes for individual user's game data such as user ID, game scores, and top score where the user-ID is the partition key. The developer must control over user access to specific game items based on their IDs. In doing so, users will only be able to obtain items that they own. Which of the following solutions must be implemented by the developer?

A. Modify the IAM Policy associated with the identity provider's role by setting the value of the dynamodb:Select condition key to the user IDs.

B. Modify the IAM Policy associated with the identity provider's role by setting the value of the dynamodb:LeadingKeys condition key to the user IDs.

C. Modify the IAM Policy associated with the identity provider's role by setting the value of the dynamodb:Attributes condition key to the user IDs.

D. Modify the IAM Policy associated with the identity provider's role by setting the value of the dynamodb:ReturnValues condition key to the user IDs.

43. A developer is tasked with enhancing the performance of an online learning platform that uses a serverless architecture. The platform relies on Amazon API Gateway to handle user requests, AWS Lambda to process quiz submissions, Amazon DynamoDB to store course progress data, and Amazon S3 to host video lectures. During peak hours, the platform experiences high latency caused by increased read operations on DynamoDB, leading to a poor user experience. Which AWS service or feature should the developer implement to optimize DynamoDB read performance and reduce latency?

A. Amazon DynamoDB Streams

B. Amazon Elastic Load Balancing

C. Amazon CloudFront

D. Amazon DynamoDB Accelerator (DAX)

44. An application that runs on a local Linux server is migrated to a Lambda function to save costs. The Lambda function shows an Unable to import module error when invoked. As the developer, how can you fix the error?

A. Import the missing modules in the Lambda code. The Lambda function will automatically install them when invoked.

B. Install the missing modules locally to your application's folder. Package the folder into a ZIP file and upload it to AWS Lambda.

C. Download the missing modules in the lib directory. Package all files under the lib directory into a ZIP file and upload it to AWS Lambda.

D. Run a Linux command inside the Lambda function to install the missing modules.

45. A developer is building an application that uses Amazon CloudFront to distribute thousands of images stored in an S3 bucket. The developer needs a fast and cost-efficient solution that will allow him to update the images immediately without waiting for the object's expiration date. Which solution meets the requirements?

A. Update the images by using versioned file names.

B. Upload the new images in the S3 bucket and wait for the objects in the edge locations to expire to reflect the changes.

C. Disable the CloudFront distribution and re-enable it to update the images in all edge locations.

D. Update the images by invalidating them from the edge caches.

46. A developer is writing an application that will download hundreds of media files. Each file must be encrypted with a unique encryption key within the application. The developer needs a cost-effective solution with low management overhead. Which of the following is the most suitable solution?

A. Use the CreateKey API command to generate a CMK for each file to encrypt them.

B. Enable the SSE-S3 for the S3 bucket. Directly store the files in the bucket.

C. Use an open-source key generator to produce a unique key. Use the key to encrypt the files.

D. Use the GenerateDataKey API command to generate a data key for each file to encrypt them. Store the encrypted data key and the file.

47. A developer needs to view the percentage of used memory and the number of TCP connections of instances inside an Auto Scaling Group. To achieve this, the developer must send the metrics to Amazon CloudWatch. Which approach provides the MOST secure way of authenticating a CloudWatch PutMetricData request?

A. Modify the existing Auto Scaling launch template to use an IAM role with the cloudwatch:PutMetricData permission for the instances.

B. Create an IAM user with programmatic access. Attach a cloudwatch:PutMetricData permission and store the access key in the instance's configuration file.

C. Create an IAM role with cloudwatch:PutMetricData permission for the new Auto Scaling launch template from which you launch instances.

D. Create an IAM user with programmatic access. Attach a cloudwatch:PutMetricData permission and update the Auto Scaling launch template to insert the access key and secret key into the instance user data.

48. A startup has recently opened an AWS account to develop a cloud-native web application. The CEO wants to improve the security of the account by implementing the best practices in managing access keys in AWS. Which actions follow the security best practices in IAM? (Select TWO.)

A. Use IAM roles for applications that need access to AWS services.

B. Regularly rotate the credentials for all the account users except for the administrator user for tracking purposes.

C. Maintain at least one access key for your AWS account root user.

D. Save the access key in your application code for convenience.

E. Delete any access keys to your AWS account root user.

49. A developer is debugging an issue in an AWS Lambda-based application. To save time searching through logs, the developer wants the function to return the corresponding log stream name of an invocation request. Which approach should the developer take with the least amount of effort?

A. Extract the invocation request id from the Event object of the handler. Call the FilterLogEvents API and use the request id to filter results.

B. Extract the log stream name from the Event object of the handler function.

C. Extract the invocation request id from the Context object of the handler function. Then, call the FilterLogEvents API and pass the request id to filter results.

D. Extract the log stream name from the Context object of the handler function.

50. A developer has been instructed to automate the creation of the snapshot of an existing Amazon EC2 instance. The engineer created a script that uses the AWS Command Line Interface (CLI) to run the necessary API call. He is getting an InvalidInstanceNotFound NotFound error whenever the script is run. What is the most likely cause of the error?

A. The image Id used in running the command for creating a snapshot is incorrect.

B. The AWS Region name used to configure the AWS CLI does not match the region where the instance lives.

C. The AWS Access Key Id used to configure the AWS CLI is invalid.

D. The AWS Region, where the programmatic access for the AWS CLI is created, does not match the region where the instance lives.

51. A Ruby developer is looking to offload some of the processing on his application to the AWS cloud without managing any servers. The subroutines must be written in Ruby, which mainly invokes API calls to an external web service. The response from the API call is parsed and stored in a MongoDB database. What should he do to develop the Lambda function in his preferred programming language?

A. Create a Lambda function with a custom runtime to use Ruby. Then include the runtime in the function's deployment package. Migrate it to a layer that you manage independently from the function.

B. Create a Lambda function using the AWS SDK for Ruby.

C. Create a Lambda function on Ruby with a custom runtime and use the AWS SDK for Ruby.

D. Create a Lambda function with a supported runtime version for Ruby.

52. A developer uses Amazon ECS to orchestrate two Docker containers. He needs to configure ECS to allow the two containers to share log data. Which configuration should the developer do?

A. Specify the containers in a single task definition and configure EFS as its volume type.

B. Use two task definitions for each container and mount an EFS volume between the tasks.

C. Use two pod specifications for each container and mount an EFS volume between the pods.

D. Specify the containers in a single pod specification and configure EFS as its volume type.

53. A developer is building a ReactJS application that will be hosted on Amazon S3. Amazon Cognito handles the registration and signing of users using the AWS Software Development Kit (SDK) for JavaScript. The JSON Web Token (JWT) received upon authentication will be stored in the browser's local storage. After signing in, the application will use the JWT as an authorizer to access API Gateway endpoints. What are the steps needed to implement the scenario above? (Select THREE.)

A. Choose and set the authentication provider for your website.

B. Create an Amazon Cognito Identity Pool.

C. On the API Gateway Console, create an authorizer using the Cognito User Pool ID.

D. Set the name of the header that will be used from the request to the Cognito Identity Pool as a token source for authorization.

E. Create an Amazon Cognito User Pool.

F. Set the name of the header that will be used from the request to the Cognito User Pool as a token source for authorization.

54. A company uses a Linux, Apache, MySQL, and PHP (LAMP) web stack to host an on-premises application for its car rental business. The manager wants to move its operation into the Cloud using Amazon Web Services. Which combination of services could be used to run the application that will require the least amount of configuration?

A. Amazon S3 and Amazon CloudFront

B. Amazon API Gateway and Amazon RDS

C. Amazon EC2 and Amazon Aurora

D. Amazon ECS and Amazon EFS

55. A company is running an Artificial Intelligence (AI) software for its automotive clients using the AWS Cloud. The software is used for identifying road obstructions for autonomous driving and predicting failure on vehicle components. The company wants to extend its usage and access based on different levels (students, professionals, and hobbyist developers) by exposing an API and monetizing it by charging based on usage. What should the company do?

A. Create three Usage Plans. Specify a quota and throttle requests according to the level of access.

B. Create three stages. Specify a quota and throttle requests according to the level of access.

C. Create three Authorizers to control API access.

D. Create three stages and enable CloudWatch metrics. Set up an alarm for each stage according to the ApiName, Method, Resource, and Stage dimensions.

56. A developer is building a serverless application that will send out a newsletter to customers using AWS Lambda. The Lambda function will be invoked at a 7-day interval. Which method will provide an automated and serverless approach to trigger the function?

A. Implement a task timer using Step Functions that will send a newsletter every week.

B. Add an environment variable named DAYS for the Lambda function and set its value to 7.

C. Configure a scheduled Amazon EventBridge (Amazon CloudWatch Events) that triggers every week to invoke the Lambda function.

D. Run a cron job in an Amazon EC2 instance that will trigger the Lambda function every week.

57. A developer is building a new feature for an application deployed on an EC2 instance in the N. Virginia region. A co-developer suggests to upload the code on Amazon S3 and use CodeDeploy to deploy the new version of the application. The deployment fails during the DownloadBundle deployment lifecycle event with the UnknownError: not opened for reading error. What is the possible cause of this?

A. The EC2 instance's IAM profile does not have the permissions to access the application code in Amazon S3.

B. Wrong configuration of the DownloadBundle lifecycle event in the AppSpec file.

C. The DownloadBundle deployment lifecycle event is not supported in the N. Virginia region.

D. Versioning is not enabled on the Amazon S3 Bucket where the application code resides.

58. A Lambda function has multiple sub-functions that are chained together to process large data synchronously. When invoked, the function tends to exceed its maximum timeout limit. This has prompted the developer to break the Lambda function into manageable coordinated states using Step Functions, enabling each sub-function to run in separate processes. Which of the following type of states should the developer use to run processes?

A. Parallel State

B. Task State

C. Pass State

D. Wait State

59. A company uses AWS CodeDeploy in their CI/CD pipeline to handle in-place deployments of their web application on EC2 instances. Recently, a new version of the application was pushed, which contained a code regression. The deployment group is configured with automatic rollback. What happens if the deployment of the new version fails?

A. CodeDeploy redeploys the last known good version of an application with a new deployment ID.

B. CodeDeploy reverts the EC2 instance to a previous AMI snapshot taken during the last successful deployment.

C. CodeDeploy pauses the current deployment, restores the last stable version from S3, and uses the deployment ID of the most recent deployment with a SUCCEEDED status.

D. CodeDeploy reroutes traffic back to the blue environment and terminates the green environment.

60. A startup is integrating an event-driven alerting tool with a third-party platform. The platform requires a publicly accessible HTTPS endpoint to receive webhook requests, which will be processed by a Lambda function. Given that the Lambda function signs each request with a secret key and includes it in the headers, the developer must ensure that the Lambda function executes the domain logic only when a request comes from a valid user. Which action would satisfy the requirement with the least amount of effort?

A. Create a Lambda function URL. Attach a resource-based policy to the function allowing anyone to invoke it only if the "lambda:CodeSigningConfigArn": "arn:aws:lambda::code-signing-config:csv" condition is present.

B. Configure API Gateway to connect with the Lambda function using a Lambda proxy integration. Create a Lambda function authorizer that validates incoming requests based on a signature provided in the HTTP headers.

C. Create a Lambda function URL. Attach a resource-based policy to the function allowing anyone to invoke it only if the "lambda:FunctionUrlAuthType": "AWS_IAM" condition is present. Write a custom authorization logic based on a signature provided in the HTTP headers.

D. Create a Lambda function URL. Attach a resource-based policy to the function allowing anyone to invoke it only if the "lambda:FunctionUrlAuthType": "NONE" condition is present. Write a custom authorization logic based on a signature provided in the HTTP headers.

61. An update was made on an API Gateway endpoint with caching enabled to improve latency requests. The developer expected to get the latest data as a response when he tested the application. However, he kept getting stale data upon trying many times. What should the developer do that will require the LEAST amount of effort? (Select TWO.)

A. Grant permission to the client to invalidate caching when there's a request using the IAM execution role.

B. Include Cache-Control: max-age=0 HTTP header on the API request.

C. Set the new endpoint as a trigger for the lambda function.

D. Include Cache-Control: no-cache HTTP header on the API request.

E. Create a new REST API endpoint and disable caching.

62. A company plans to conduct an online survey to distinguish the product from those who don't. The survey will be processed by Step Functions which comprises four states that will manage the application logic and error handling of the state machine. It is required to aggregate all the data that comes through the nodes of the state machine. If any of the process fails. What should the company do to meet the requirements?

A. Include a Catch field in the state machine definition to capture the errors. Then, use ResultPath to include each node's input data with its output.

B. Include a Catch field in the state machine definition to capture the errors. Then, use ItemsPath to include each node's input data with its output.

C. Include a Parameters field in the state machine definition to capture the errors. Then, use ResultPath to include each node's input data with its output.

D. Include a Parameters field in the state machine definition to capture the errors. Then, use ItemsPath to include each node's input data with its output.